



FIRST SEMESTER 2026-2027

Course Handout Part-II

Date: 02-08-2026

In addition to part -I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : CE G527

Course Title : CONSTRUCTION MANAGEMENT

Instructor-in-charge : ANSHUMAN

Room No. : 1232 (FD-I) 12 Noon – 12.50 PM

1 Course Description: The course deals with planning techniques, engineering economy, valuation including estimation & costing, system design, construction methodologies ranging from small buildings to large infrastructure projects, construction technologies including recent advancements. Relevant Indian Standard codes of practices will be given emphasis.

2 (a): Scope of the Course:

The scope of the course is to know the various aspects of construction planning and management right from the knowledge of various activities till the estimation, costing, valuation, tendering etc., Though this course covers wider infrastructure projects, the primary focus will be on building and road infrastructure.

2(b): Objectives of the course:

- To understand the different construction projects and basic objective of project management [**Module - I**]
- To understand the project management techniques (CPM and PERT) [**Module - II**]
- To analyze the various requirements of manpower and other resources [**Module-III**]
- To analyze the various control methodologies keeping in line with project requirements [**Module-IV**]
- To get exposed to construction contracts and tenders [**Module-V**]
- To estimate the quantity and rate of different building components (including materials) [**Module VI**]
- To discuss the various aspects of valuation of properties [**Module VII**]
- To highlights various aspects of construction quality and safety with case studies [**Module VIII**]
- To understand the best practices of construction management [**Module IX**]





3.Key Learning outcomes:

The students will be trained to

- Demonstrate a systematic understanding of the function of project management in property, construction, and the project lifecycle; and of the concepts, theories, principles, and techniques associated with construction project management at strategic and operational level.
- Critically apply established techniques of analysis and enquiry to construction project management and use judgement to evaluate a range of solutions.
- Provide appropriate construction management advice to organizations in order to meet objectives while meeting required standards of ethics and governance.
- Demonstrate the skills to manage procedures and processes during the construction project lifecycle.
- Understand the legal implications of contract, common, and regulatory law to manage a construction project

4. Text Book:

T1. Kumar Neeraj Jha, Construction Project Management, Pearson India publications

T2. K.K. Chitkara Construction Project Management, Tata McGraw Hill publications.

4. Reference Books:

R1. David Pratt, Fundamentals of Construction Estimates, 3rd edition, Cengage publications.

R2 Albert Lester, Project Management, Planning and Control, 7th edition, BH (Elsevier) publications

R3. Gupta R. (2015) "Construction Planning & Technology", CBS Publishers, New Delhi.

5. Course Plan

Mod . No.	Module	Topics to be covered	Luct ures	Learning outcomes
1	Introduction to Construction activities and management	• Construction Industry • Project categories • Project classification • Project Life cycle • Project participants • Project Management and Functional Management	1-3	To get overall exposure about construction management principles
2	Project Management with CPM and PERT	• Real time project networking of big infrastructure projects. • Construction of a network	4-8	To understand various construction activities of big infrastructure projects and converting into a network
3	Resource Planning including time planning	• Planning construction manpower • Planning construction materials • Project construction equipment including selection of equipment	9-13	To evaluate the comprehensive details of the various key planning operations (manpower, materials, equipment)





BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
Instruction Division

		Construction costs & construction budget		
4	Planning control system	- Project control methodology - Resources productivity control - Project cost control - Project Time control - Construction accounts management	14-16	To idealize the correct control methodology to be adopted for a project
5	Tenders, contracts, and estimates	- Types of contracts. - Contracts terms and conditions, special clauses, terminologies - Dispute cases, termination, liquidation, incentive, JOC, - Discussion on actual tender document	17-24	To get the adequate confidence in preparing tender documents including overview of costs.
6	Estimation of various building components	- Cost aspects of various projects (Case studies), pricing of various activities (earthwork excavation, equipment etc) - Types of estimates - Activity wise cost estimates (case studies)	25-31	To get the hold on estimation, costing, pricing of major components/activity.
7	Construction Valuation	- Valuation, engineering economy - Discussion on guidelines on valuation of immovable properties (standard practice, specifications, methods) - Cost-benefit analysis	31-35	To comprehend the various guidelines and specifications in the valuation of buildings
8	Construction quality and safety	- Construction quality - Inspection, Quality Assurance - Cost of quality and ISO standards - Causes of low labor productivity - Accident Causation Theories - Safety and Health Management system - Research results in safety management	35-38	To understand the quality and safety aspects of construction
9	Best practices of Construction Management	- Construction companies' best practices. - Introduction to Building Information Modeling (BIM) - Failures of projects	39-40	To acknowledge the best practices adopted by construction giants.
		Total	40	





6. Evaluation Scheme:

Component	Duration	Weight	Date & Time	Remarks
Mid-Test	60 min	30%	06/10 AN2	CB
Project	Semester	20%		OB/CB
Assignments	Semester	10%		OB
Comprehensive Examination	180 min	40%	05/12 AN	CB/OB

7. Reading assignments will be given as and when necessary

8. Notices: Notices will be sent through your BITS mail only.

Instructor-in-charge

